João Dimas

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Education

Washington University in St. Louis
PhD in Finance, Olin Business School 2022–2026
Advisor: Todd Gormley
Dissertation: The Economics of Technical Debt: Venture Capital, Cybersecurity, and Engineering Quality

Washington University in St. Louis
Doctoral studies in Economics; transferred to the PhD in Finance program. 2020–2022

Université Paris I Panthéon-Sorbonne
MSc in Financial Economics 2018

Universidade FUMEC
B.Sc. in Computer Science 2009

Research Fields

Corporate finance and the economics of technology (AI, software quality, and cybersecurity).

Job Market Paper

Venture Capital and Engineering Quality: Evidence from Software Startups

with L. Pnevmatikos (third-year PhD student)

Buchan Prize, 1st place (best PhD student-written paper).

We investigate a novel channel of venture capital value creation: the practices through which startups build and maintain their technology. We construct a new panel of 569 firms using more than 60,000 full codebase snapshots. We compare changes in engineering quality following first VC investment with changes at similar firms that remain without VC backing. We also construct a dollar-denominated measure of technical debt, the estimated maintenance liability embedded in software. Relative to codebase size, serious deviations from coding best practices decline by 15% and technical debt by 12%. These improvements accompany substantial codebase growth. The decline in the frequency of serious coding problems persists after matching on pre-financing growth, and quality gains extend to firms already generating revenue. In contrast, a placebo test around later financing rounds shows no evidence of quality improvement. We also analyze 5.6 million code contributions (commits). After first VC investment, teams more often document collaboration and copyright compliance, the share of weekend contributions declines, and existing engineers become less likely to introduce bugs. The findings suggest that VC professionalization extends to the engineering practices through which teams build and maintain their products.

Working Papers

AI and Technical Debt: Evidence from 3 Billion Lines of Code

with E. Lu and L. Pnevmatikos

AI raises worker productivity and lets firms build knowledge assets, such as the software they run on, faster. Low-quality AI output, however, may embed these assets with hidden liabilities that firms must later verify, fix, and maintain. Software lets us inspect the asset itself and measure these liabilities directly. We identify firm-affiliated public GitHub projects and, every month, use an industry-standard tool to scan each project's full codebase for maintainability issues, bugs, and security vulnerabilities. This amounts to processing over 3.6 billion lines of code. We convert the tool's estimated repair time into a dollar measure of technical debt. We compare projects in which AI use first appears with similar projects whose AI use is detected later or remains undetected. During the first seven months of detected AI use, codebases grow on average 6.1% more than those of comparable projects. Contrary to concerns about "AI slop," total issues per line of code decline by 1.3%, and technical debt per line by 0.7%. Bugs per line also decline, and security vulnerabilities and critical issues, the most severe ones, show no statistically significant change. The quality gains are modest, but at the very least they provide evidence against a deterioration in the quality of the average line. Detected AI use is thus linked with faster growth in firms' software assets.

Patch Now, Pay Later? Cybersecurity Vulnerabilities and Technical Debt

with L. Pnevmatikos

We study how cybersecurity vulnerabilities affect firm performance. We analyze 551 open-core software firms from 2010 to 2025, comparing firms affected by vulnerability discoveries with unaffected firms. These discoveries concern third-party dependency packages, making their timing plausibly exogenous to the firms using them. Exposed firms experience approximately 10% slower headcount growth relative to their pre-treatment average and a 1.7 percentage point decline in acquisition probability. Commit-level evidence reveals a crowding-out mechanism: the share of non-remediation development activity falls by 2.78 percentage points, and fix-related effort persists for twelve months, displacing feature development. Technical debt and cognitive complexity grow approximately 20 and 25 percentage points faster, respectively. Together, the findings suggest that cybersecurity vulnerabilities can disrupt firm growth and place persistent demands on engineering resources.

Work in Progress

Acquisition Prospects and Startup Strategy

with A. Jain and L. Pnevmatikos

We examine how a decline in acquisition prospects changes startups' product development and hiring, and whether it encourages investment in more durable standalone businesses.

Hedge or Speculation? Evidence from Brazil

with Gustavo Cortes, Sergio Leao, and Alexandre Pecora

We study how Brazilian non-financial firms adjust currency-derivative positions in response to market conditions and the past performance of their positions, distinguishing hedging from speculation.

Industry Papers

Toh, W. Q., W. Zheng, Y. Wang, and J. Dimas (2025). A White Box Framework for Estimating Long-Term Experimental Impact via Short-Term Metrics. ACM KDD 4th End to End Customer Journey Optimization Workshop.

We develop a framework for estimating long-term experimental effects from short-term metrics while making the underlying causal mechanisms explicit. The framework connects short-term measures to a long-term outcome through its key drivers, validates estimates using both pseudo-experiments and actual experiments, and addresses implementation on an experimentation platform.

Honors

Buchan Prize, 1st place (best PhD student-written paper), with L. Pnevmatikos.

Conferences

12th Annual USC Marshall PhD Conference in Finance June 15, 2026
Job market paper, presented by L. Pnevmatikos.

Research and Professional Experience

Roblox
PhD Data Scientist Intern, Experimentation Platform 2025

- Developed methodology for estimating 365-day customer lifetime value treatment effects from 14-day experimental outcomes.
- Estimated engagement–monetization trade-offs and designed validation against realized long-run outcomes to inform experimentation-platform decisions.

Central Bank of Brazil (Banco Central do Brasil)
Economist, Research Department 2018–2021

- Contributed to research on inflation dynamics and monetary-policy transmission, producing forecasts and alternative macroeconomic scenarios to support policy decisions.
- Researched credit markets, banking efficiency, financial stability, and their effects on the real economy.
- Developed macroeconomic modeling and forecasting tools and reproducible empirical workflows; contributed analytical inputs to the Inflation Report and policy briefings.

Software Engineer, IT Department 2012–2016

CI&T
Software Engineer and Technical Lead 2008–2012

CEMIG
Software Engineer 2007–2008

Teaching Experience

Washington University in St. Louis

Teaching Assistant	2021–2025

PhD-Level Courses

Applied Econometrics	Fall 2021
Empirical Methods in Finance	Spring 2024
Introduction to Asset Pricing	Fall 2024

Other Graduate Courses

Advanced Corporate Finance I — Valuation	Spring 2024
Advanced Corporate Finance II — Financing	Fall 2022; Spring 2024
Derivative Securities	Fall 2023, 2024, 2025

Undergraduate Courses

Advanced Financial Management	Spring 2024, 2025
Computational Macroeconomics	Spring 2022
Intermediate Microeconomic Theory	Spring 2022
Options, Futures and Derivative Securities	Fall 2025

References

Olin Business School, Washington University in St. Louis

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